

ATBASAR, Kazakhstan

WMO: 350780

Lat: 51.8419N

Lon: 68.3636E

Elev: 304

StdP: 97.73

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-30.8	-36.5	0.1	-33.5	-33.6	0.2	-30.6	16.5	-10.2	14.5	-9.1	2.9	200	0.710

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.4	(c) 32.3	(d) 17.5	(e) 30.4	(f) 17.1	(g) 28.5	(h) 16.7	(i) 19.6	(j) 27.4	(k) 18.8	(l) 26.8	(m) 18.0	(n) 25.7	(o) 5.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.8	21.8	16.3	12.0	21.2	15.3	11.3	20.6	57.4	27.3	54.8	26.9	52.1	25.5	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.0	(b) 12.0	(c) 10.3	(e) -37.0	(f) 35.5	(g) 3.1	(h) 2.1	(i) -39.3	(j) 37.0	(k) -41.1	(l) 38.2	(m) -42.9	(n) 39.4	(o) -45.2	(p) 40.9		
(4) 14.0	(5) 12.0	(6) 10.3	(7) -36.6	(8) 21.1	(9) 3.3	(10) 1.2	(11) -39.0	(12) 22.0	(13) -40.9	(14) 22.7	(15) -42.8	(16) 23.4	(17) -45.2	(18) 24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-17.5	-16.0	-8.2	5.3	13.1	18.2	19.6	18.3	12.0	4.3	-6.3	-13.9
	DBStd	14.82	7.83	6.97	6.90	6.35	4.81	4.17	3.57	4.32	4.99	5.15	7.76	7.72
	HDD10.0	3791	852	729	563	165	25	2	0	1	34	188	491	741
	HDD18.3	5969	1111	962	821	392	172	52	28	56	199	436	741	999
	CDD10.0	1048	0	0	0	23	120	248	296	257	93	11	0	0
	CDD18.3	183	0	0	0	8	48	66	53	8	0	0	0	0
	CDH23.3	2434	0	0	0	10	173	652	771	692	133	3	0	0
	CDH26.7	837	0	0	0	1	38	226	261	266	45	0	0	0
	WSAvg	4.7	4.7	5.2	5.3	5.0	5.0	4.5	4.2	4.0	4.2	4.5	4.9	5.1
	PrecAvg	305	22	15	15	17	30	38	48	32	23	25	21	22
(15) Precipitation	PrecMax	454	83	62	45	47	70	154	152	92	58	87	56	59
	PrecMin	156	1	0	1	0	3	2	0	2	0	1	0	3
	PrecStd	59	14	11	9	12	16	28	34	22	14	18	14	13
	PrecStd	59	14	11	9	12	16	28	34	22	14	18	14	13
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	0.1	8.2	24.7	30.4	33.9	34.4	34.7	31.7	22.4	9.4	0.6
		MCWB	-0.2	-0.3	5.0	12.8	14.9	17.9	18.9	17.8	15.8	10.6	6.1	-0.1
	2%	DB	-1.6	-1.4	3.5	21.3	27.4	31.3	31.6	32.0	27.3	18.6	6.3	-0.3
		MCWB	-2.0	-1.8	2.1	11.2	14.5	17.4	18.0	17.2	14.0	9.8	4.5	-0.7
	5%	DB	-3.7	-3.2	2.1	18.6	25.1	29.4	29.6	29.7	23.7	15.5	4.1	-1.8
		MCWB	-4.1	-3.7	1.1	10.4	13.7	17.0	17.6	17.0	13.0	8.3	2.7	-2.3
	10%	DB	-5.5	-5.1	0.9	15.5	23.1	27.2	27.6	27.4	20.9	12.5	2.2	-3.5
		MCWB	-6.0	-5.5	0.1	8.9	12.9	16.3	17.3	16.4	11.9	7.0	1.2	-4.0
	0.4%	WB	-0.1	-0.3	5.4	14.2	17.4	20.2	20.7	19.8	17.1	12.1	7.0	0.2
		MCDB	0.1	0.0	8.0	21.3	25.1	28.4	28.2	28.4	27.6	19.1	8.2	0.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-1.9	-1.8	2.3	12.4	15.6	19.1	19.9	18.9	15.2	10.5	4.7	-0.7
		MCDB	-1.5	-1.5	3.2	19.0	23.9	27.5	27.1	27.9	23.9	16.7	6.3	-0.3
	5%	WB	-4.0	-3.7	1.2	10.9	14.6	18.1	19.2	18.1	13.8	9.0	2.9	-2.2
		MCDB	-3.7	-3.2	1.9	17.2	22.8	26.4	26.6	26.7	22.2	14.2	4.0	-1.9
	10%	WB	-5.9	-5.5	0.2	9.3	13.7	17.2	18.4	17.2	12.6	7.5	1.2	-4.0
		MCDB	-5.4	-5.0	0.9	15.2	21.7	25.2	25.4	25.3	19.5	12.1	2.2	-3.5
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.7	8.8	11.1	13.4	12.9	12.4	13.1	12.6	10.1	7.1	7.7
		MCDBR	8.0	7.7	6.7	16.2	16.7	16.1	15.5	16.8	16.8	15.7	8.0	7.8
		MCWBR	8.0	7.7	5.7	7.8	7.3	6.1	5.5	6.1	7.1	8.5	6.3	7.7
	5% WB	MCDBR	8.0	7.8	6.4	14.7	14.4	13.6	12.7	14.4	14.8	13.6	7.4	7.6
		MCWBR	8.1	7.8	5.6	7.8	7.1	5.9	5.2	6.0	7.0	8.1	6.2	7.6
(40) Clear-Sky Solar Irradiance	taub		0.290	0.320	0.373	0.406	0.400	0.404	0.420	0.411	0.359	0.344	0.322	0.289
	taud		2.065	2.020	1.961	2.133	2.235	2.295	2.245	2.248	2.343	2.291	2.149	2.059
	Ebn at Noon		678	767	795	825	852	851	828	814	820	748	634	602
	Edh at Noon		79	111	144	138	131	125	129	122	98	85	73	67
	RadAvg		1.34	2.47	3.85	4.77	5.92	6.27	5.78	5.06	3.73	2.04	0.95	0.88
(45) All-Sky Solar Radiation	RadStd		0.10	0.12	0.41	0.45	0.52	0.52	0.45	0.37	0.28	0.27	0.11	0.15

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page